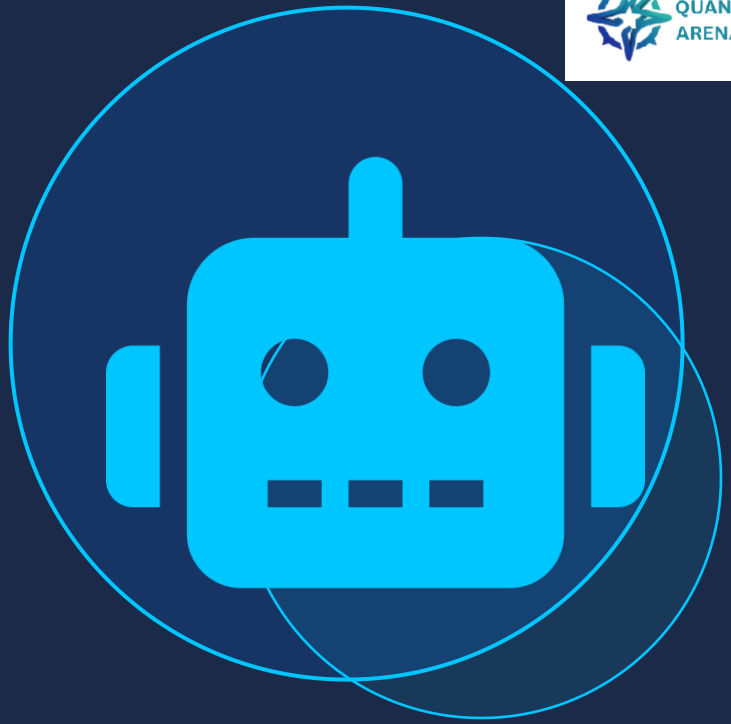
AI CHATBOT FOR ACADEMIC SUPPORT



Problem Statement: AI Chatbot for Academic Support

Theme: NLP + AI

PS Category: Software

Team Name: **sensors squad**

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IDEA TITLE

QUANTUM ARENA 1.0

PROPOSED SOLUTION

1. Intelligent NLP-powered chatbot integrated into campus portals
2. Understands natural language queries from students in real-time
3. Provides instant, accurate answers on exams, timetables & placements
4. Context-aware memory to maintain conversation continuity

✦ INNOVATION

🎯 **Multi-domain NLP**

Handles exams, fees, placements

🧠 **Context Memory**

Recalls past queries in session

⚡ **Sub-second Response**

Instant reply via optimized API

🔒 **Secure Auth**

Student-specific data access

📱 **Omni-channel**

Web, mobile & WhatsApp support

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TECHNICAL APPROACH

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SYSTEM ARCHITECTURE FLOW

User Input
Web / Mobile / WhatsApp

NLP Engine
Intent & Entity Recognition

Backend API
FastAPI / Node.js Business Logic

Database
MongoDB + Redis Cache Layer

Response
Formatted Answer with Context

TECH STACK

- **NLP/AI:** Dialogflow / Rasa / OpenAI API
- **Backend:** FastAPI (Python) + Node.js
- **Database:** MongoDB + Redis Cache
- **Frontend:** React.js / Flutter (Mobile)

IMPLEMENTATION PHASES

01 Phase 1: NLP Training & Intent Mapping

02 Phase 2: Backend API + DB Integration

03 Phase 3: UI Development & Auth Setup

04 Phase 4: Testing, Feedback & Deployment

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FEASIBILITY AND VIABILITY

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✅ FEASIBILITY ANALYSIS

- Pre-trained NLP APIs (OpenAI, Dialogflow) readily available
- Lightweight backend deployable on cloud (AWS / GCP free tier)
- MongoDB Atlas free cluster for data storage
- React.js frontend buildable within 2-week sprint
- MVP achievable within hackathon timeline

⚠️ CHALLENGES & RISKS

⚠ Accuracy of NLP responses

→ Fine-tune with college-specific data corpus

⚠ Data privacy of students

→ OAuth 2.0 auth + encrypted DB fields

⚠ Scale under peak load

→ Redis caching + load balancer

⚠ Admin content updates

→ Admin CMS portal for easy edits

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IMPACT AND BENEFITS

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80%
Reduction in admin query workload

24/7
Student support availability

<1s
Average response time

5000+
Students benefited per campus

🎓 Social Impact

- Equal access to academic info for all students
- Reduces digital divide in campus services
- Empowers first-gen students with instant guidance

💰 Economic Impact

- Saves 2-3 staff hours/day on repetitive queries
- Low-cost cloud deployment (< ₹500/month)
- Scalable across multiple college campuses

🌍 Institutional Impact

- Boosts institutional efficiency and modernization
- Real-time analytics on student concerns
- Reduces student dropout due to info gaps

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RESEARCH AND REFERENCES

QUANTUM ARENA 1.0

RESEARCH REFERENCES

[1] Conversational AI in Education

IEEE Xplore (2023) — NLP-driven chatbots in university student support https://ieeexplore.ieee.org

[2] Rasa Open Source Framework

Official Docs — Building contextual AI assistants with Rasa NLU https://rasa.com/docs

[3] OpenAI GPT API

OpenAI Platform — Using GPT-4 for domain-specific Q&A systems https://platform.openai.com/docs

[4] Dialogflow CX — Google Cloud

Google Developers — Intent classification & entity recognition at scale https://cloud.google.com/dialogflow

[5] MongoDB Atlas for Academic Apps

MongoDB Blog — Scalable document storage for student data https://www.mongodb.com/atlas

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REQUIREMENTS

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HARDWARE REQUIREMENTS

Component	Cost
RAM: 4GB or more	free
Storage: 1GB	free
OS: Windows 10/11 / Linux / Mac	free
Processor: i3	free
smartphone for mobile testing	free

SOFTWARE REQUIREMENTS

Software	Cost
Python	Free
Flask	Free
SQLite	Free
VS Code	free
HTML/CSS/JS	free
Local hosting	free

Web page link:
http://192.168.1.110:5000

TOTAL ESTIMATED COST: ~Completely free